MSC THESIS DEFENCE

Measuring and Prioritising Technical Debt in Mission-Critical Trading Systems

A TOPSIS framework on live ITSM and CMDB data from a regulated Swiss trading institution

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MOTIVATION

The problem lives outside the code

- In vendor-dominated trading IT, source code is inaccessible, so the dominant tools (SonarQube, SQALE, CAST) are structurally inapplicable
- The debt that matters sits in infrastructure lifecycle exposure, patch delay and operational instability, which code-centric tools cannot observe
- Its symptoms, however, are recorded daily in ITSM and CMDB systems

−16%
gross return on assets per 10% more technical debt in COTS platforms (Banker et al., 2020)

4.9× / 9.1×
higher odds of compromise after one / three months of patch delay (Di Tizio et al., 2023)

−84%
incidents after targeted architectural debt repayment (de Toledo et al., 2021)

TAKEAWAY
The costliest debt in this environment is invisible to the standard tools, but its symptoms are already being recorded.

The gap and the research questions

Across 44 primary studies on TD prioritisation, quantitative multi-criteria approaches are uncommon and industrial validation is rare (Lenarduzzi et al., 2021). **No reviewed study combines** service-level indicators, multi-criteria prioritisation, live operational data and a trading IT context.

RQ1	How can technical debt in mission-critical trading systems be prioritised quantitatively at the service level using operational data?
RQ2	To what extent do operational service management metrics serve as valid proxies for technical debt severity?
RQ3	How stable are the resulting rankings when criterion weights are perturbed?

Five indicators, one auditable instrument

DSR (Hevner et al., 2004) · single-organisation embedded case study · unit of analysis: the IT service as a configuration item

INDICATOR	SOURCE	DEBT DIMENSION (TABLE 2.3)
Incident frequency	ITSM	Architectural
Mean time to restore (MTTR)	ITSM	Infrastructure / platform
Change failure rate (CFR)	ITSM	Process
Patch recency	CMDB	Vulnerability
Unsupported component months	CMDB	Infrastructure lifecycle

Dimension assignments are this study's literature-grounded synthesis. Documentation, test coverage and coupling indicators were excluded on data availability, not relevance.

From records to ranking

Pipeline

1. Extract indicators from live ITSM / CMDB records
2. Vector normalisation (preserves extreme observations)
3. Weighting: equal (0.20 each) and differential (0.30 MTTR & patch recency)
4. TOPSIS closeness coefficient $C_i \in [0, 1]$
5. Rank: lowest C_i = highest debt priority

Why TOPSIS

- Handles mixed cost/benefit criteria and different scales in one procedure
- Produces a full ranking, unlike VIKOR's compromise set or AHP alone
- Precedent: Albarak & Bahsoon (2022) for TD, but on controlled data
- **Method choice treated as a robustness question, tested against SAW and VIKOR**

TAKEAWAY

Differential weights are a declared practitioner assumption, and the evaluation is designed so nothing depends on them.

The data, and its honest limits

Dataset

- Portfolio of 108 services; 53 small vendor GUIs excluded; 6 selected for demonstration on data completeness over the full window
- Jan 2024 – Dec 2025, production records only, anonymised at source
- Quality filters: alarm-storm consolidation, <0.1 h auto-closures removed, 98th percentile MTTR cap

Declared limitations

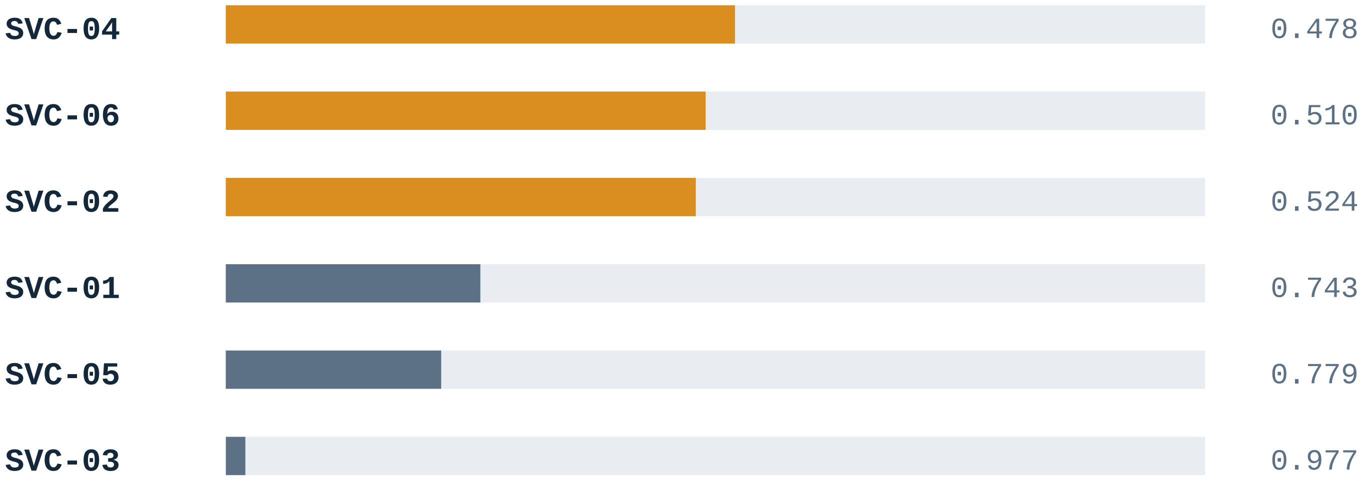
- Recording practices vary by team → scores are ordinal signals, not measurements
- CFR shares its numerator with incident frequency (bounded by sensitivity analysis)
- Demonstration set, not a portfolio ranking; the contribution is the instrument and its evaluation logic

TAKEAWAY

Declared limitations are part of the audit trail: each is stated in the thesis, and the small-n consequence is handled statistically in the evaluation.

RESULTS · 2025, EQUAL WEIGHTS

The priority ranking



SVC-04 · rank 1

574 incidents/yr, MTTR 41.9 h, CFR 3.38, yet a clean lifecycle record → architectural, platform & process debt

SVC-02 · rank 3

Only 30 incidents/yr, but 1,221 unsupported component months → the debt an incident count cannot see

SVC-03 · rank 6

The portfolio's reference for a well-maintained service

TAKEAWAY

Each position maps onto a distinct debt dimension from Table 2.3: the practical-validity argument for RQ2.

Five strands, no single point of trust

Stability

Weight & temporal variation, Kendall's τ, exact p over 720 permutations

$\tau = 0.73\text{–}1.00$

Weight space

20,000 Dirichlet-sampled weightings, rank-acceptability analysis

`extremes robust`

Convergent

Re-ranked with SAW & VIKOR

$\tau = 0.60\text{–}0.87$

Discriminant

Against an incident-only baseline

$\tau = 0.60$ / 0.20

Structural

Leave-one-out rank reversal; absolute-mode variant

`reversal-free by construction`

TAKEAWAY

No external ground truth exists for non-code debt, so validity is triangulated: robustness, method-independence and discriminant value are established; criterion validity is explicitly not claimed.

RQ3 · STABILITY

Robust where it matters

Discrete configurations

- Weight sensitivity 2025: $\tau = 0.867$ (one adjacent swap)
- Temporal 2025 vs 2024: $\tau = 0.867$ / 1.000
- Weakest: 2024 weights, $\tau = 0.733$, tightly grouped mid-field profiles
- **Top and bottom positions identical in every run**

FULL WEIGHT SPACE · 20,000 DRAWS

99.1%

of all admissible weightings place SVC-03 last; the high-priority cluster {SVC-04, SVC-06, SVC-02} holds the top three in the large majority. Sensitivity is confined to the middle ranks.

TAKEAWAY

Cross-year movement is read as descriptive context, not a quality criterion: services genuinely evolve, and a faithful ranking should follow them.

DISCRIMINANT VALUE

Does it earn its complexity?

Against a ranking by incident frequency alone: **$\tau = 0.600$** (2025), **$\tau = 0.200$** (2024).

SVC-02 rises

Infrastructure lifecycle debt (1,221 unsupported component months, patch recency of 59.93, second lowest) is invisible to an incident count. This is precisely the signal the framework was built to surface.

SVC-05 falls

High incident volume (171/yr) offset by full patch recency and zero unsupported components: noisy, but structurally sound.

TAKEAWAY

The divergence is not noise; it concentrates exactly where multi-criteria measurement should add information over a single signal (Albarak & Bahsoon, 2022).

Where the methods disagree, and why that is a finding

All methods agree on the portfolio halves and on SVC-03 last, in every year and configuration. They disagree on one thing: **SVC-06.**

	SAW / VIKOR	TOPSIS ABSOLUTE MODE
2024 rank of SVC-06	1st (VIKOR regret at maximum)	5th ($C^* = 0.853$)
2025 rank of SVC-06	1st	1st ($C^* = 0.609$)
Agreement, 2024	—	$\tau = 0.20 / 0.33$
Agreement, 2025	—	$\tau = 1.00$

MECHANISM

Sample-relative methods punish a worst-in-portfolio value however moderate it is absolutely. Absolute-mode TOPSIS measures distance to fixed anchors, so squared aggregation lets strong criteria dilute one concentrated weakness.

TAKEAWAY

SVC-06's 2024 debt was concentrated; by 2025 it had broadened in absolute magnitude (VIKOR utility 0.491 → 0.559, regret still capped) and every method converged. Two lenses, one coherent story.

METHODOLOGICAL CONTRIBUTION

Division of labour, not a winner

Standard TOPSIS + SAW / VIKOR

- Primary prioritisation signal
- Non-compensatory regret catches concentrated deficiencies, often the signal that matters most in debt work

Absolute-mode TOPSIS

- Rank-reversal free by construction: fixed bounds and anchors (García-Cascales & Lamata 2012; Yang 2020)
- Scores comparable across observation periods, uniquely among the methods used
- Complementary perspective; also powers the Monte Carlo cleanly

TAKEAWAY

Whether SVC-04 or SVC-06 deserves the first remediation slot is a judgement about whether debt dimensions are compensatory. The framework informs that judgement; it does not settle it, and saying so is a feature.

Anticipated questions

Q **Only six services: what can that prove?**

The contribution is the instrument and evaluation logic, not a portfolio ranking. Selection was by data completeness; p-values are exact over all 720 permutations and read as effect sizes; portfolio-wide deployment is stated future work.

Q **You built and evaluated on the same data. Circular?**

Declared openly: the evaluation establishes robustness, method-independence and discriminant value, not criterion validity, because no independent measure of non-code debt exists. External validation is the primary future-work avenue.

Q **Are DORA metrics really debt proxies?**

Treated as the central testable hypothesis, never a premise. Forsgren et al. validate the metrics; Nord et al. ground the diagnostic reading; the rankings match literature-predicted profiles. Longitudinal tracking would move it from plausible to evidenced.

Anticipated questions, continued

Q Aren't the differential weights arbitrary?

Yes, and they are declared as a practitioner assumption. That is why the evaluation does not depend on them: 20,000 Dirichlet-sampled weightings show the extreme priorities hold under any admissible weighting.

Q Could this be misused to judge teams?

Governance scope was agreed with the Head of Trading IT at the outset: decision-support for portfolio conversations, not a binding instrument. High-debt profiles typically reflect historical design decisions, not current custodians (Ahmad et al., 2026). No team, vendor or individual identifiers exist in the dataset.

TAKEAWAY

The framework measures debt indirectly and expresses the result as a relative priority: a quantitative, reproducible ordering, not a claim to have quantified the debt itself.

Contributions

1. **First application:** (to the best of current knowledge) of TOPSIS to live ITSM/CMDB records for TD prioritisation in financial trading IT, closing the gap Lenarduzzi et al. identify field-wide

2. **An operationally defined indicator set:** with explicit literature mapping to debt dimensions, plus evidence of its behaviour under a four-strand evaluation

3. **A methodological finding of independent interest:** sample-relative and absolutely anchored methods respond differently to concentrated debt profiles, consequential for the top of a ranking

TAKEAWAY

Practically: an auditable priority ordering from data organisations already hold, aligned with FINMA 2023/1 operational-resilience expectations, surfacing exactly the debt current governance misses.

CLOSE

The framework does not measure technical debt.
It orders services by the **operational shadow debt casts**, and within stated limits it does so reproducibly, defensibly and in a form a regulated institution can act on.

Thank you. I welcome your questions.

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